



Mustafa Arif

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SUMMARY

Android Architect | Senior Android Developer | Mobile Security Expert | Kotlin | Jetpack Compose | BLE | Play Integrity API | Secure App Development

Security-focused Android Developer with 12+ years of software engineering experience and 8+ years in Android development, delivering scalable, secure, and high-performance mobile apps in e-commerce (eBay) and regulated healthcare (Medtronic) domains.

Specialized in mobile app security architecture, including biometric authentication, Trusted Execution Environment (TEE), Android Keystore, and cryptographic protocols such as ECC, ECDSA, ECIES, and ECDH. Deep experience with Play Integrity API, SafetyNet, secure token handling, root/jailbreak detection, CVSS scoring, and threat modeling.

Expert in Modern Android Development (MAD) with Jetpack Compose, Kotlin Coroutines, MVVM, Hilt, and modular clean architecture.

Delivered multiple apps with advanced BLE communication, custom GATT services, and L2CAP tuning for medical-grade IoT integrations.

Proven ability to lead cross-functional, remote teams, and implement CI/CD pipelines, automated testing, and compliance with HIPAA, FDA, and enterprise security policies (e.g., Intune).

Passionate about building secure, testable, and privacy-first Android applications on a scale.

SKILLS

Languages: Kotlin, Java, Swift, Objective-C, Dart, Python, C++

Android Frameworks: Jetpack Compose, MVVM, Hilt, Coroutines, Flow, Navigation, Room, WorkManager

Mobile Security: Play Integrity API, SafetyNet, Android Keystore, TEE, ECC, ECDSA, ECIES, Biometric Authentication, CVSS, Root Detection

BLE/IoT: Bluetooth Low Energy, GATT, L2CAP, Custom Services, Secure Communication Protocols

Testing & QA: Espresso, JUnit, Mockito, Firebase Test Lab, Compose UI Testing

CI/CD & DevOps: GitHub Actions, Jenkins, Firebase App Distribution, Fastlane

Cross-Platform & iOS: Flutter, Capacitor, Ionic, Swift, Objective-C

Analytics & Tools: Firebase Crashlytics, Adobe Analytics, Android Studio, ADB, Logcat

Architecture: Clean Architecture, Modularization, SOLID, MVVM, MVI

EXPERIENCE

ANDROID APP ARCHITECT & MOBILE SECURITY EXPERT

April '22– Present

eBay Inc – Sanjose, CA (Remote)

Project:

eBay: The shopping marketplace:

High-scale Android e-commerce application enabling millions of global users to securely buy and sell across fashion, electronics, collectibles, and more.

- ➔ Architected the Android app using Kotlin, Jetpack Compose, MVVM, Clean Architecture, and modularization, enabling long-term scalability and testability.
- ➔ Implemented Time-based One-Time Passwords (TOTP) using open standard algorithms (RFC 6238), supporting multi-factor authentication (MFA) for high-risk user actions.
- ➔ Implemented advanced biometric authentication flows using Android Keystore, Secure Enclave, and TEE-backed ECC KeyPairs to secure critical user actions.
- ➔ Designed and deployed SSO (Single Sign-On) authentication across the mobile app, integrating with identity providers and platform APIs to streamline secure access.
- ➔ Integrated FIDO2/WebAuthn-based Passkey authentication, enabling passwordless and phishing-resistant login across devices with TEE-based credential binding.
- ➔ Developed and maintained a PCI-compliant SDK for secure in-app payments, supporting encrypted token flows, multi-tenant use, and signature-based authorization.
- ➔ Engineered anti-bot protection architecture, utilizing Play Integrity API, App Integrity API, device attestation, and rate-limited access tokens.
- ➔ Delivered NFC-based badging integration using HID Global systems, supporting secure internal mobile access with encrypted credential exchange.
- ➔ Implemented mobile ML security hardening, ensuring encrypted model assets, runtime validation, and protection against unauthorized inference access.
- ➔ Managed third-party SDK governance using JFrog Artifactory, including static analysis, vulnerability scans (Snyk/OWASP), and security documentation compliance.
- ➔ Authored and reviewed CVSS-based threat models, and led risk assessments across all new features as part of a secure SDLC.
- ➔ Acted as feature security architect, reviewing every major release for architectural integrity, secure design, and regulatory compliance.
- ➔ Contributed to Intune MDM integration, mobile device policy enforcement, and enterprise readiness alignment.
- ➔ Developed and maintained Espresso UI test automation, covering login, payment, SSO, and device attestation flows.
- ➔ Defined and enforced security policies covering:
 - Local data storage
 - WebView configurations (secure JS interface, mixed content policy, file scheme restrictions)
 - On-device cryptography (key generation, key use auditing, secure protocols)
 - ML model lifecycle and inference security

ANDROID APP ARCHITECT & MOBILE SECURITY EXPERT

June '19– April '22

Medtronic Inc – Northridge, CA

Project:

Minimed+® App (Pump Control App):

The AHCL+ ® mobile app connects to insulin pump via BLE to display pump therapy, remote bolus, and continuous glucose monitor (CGM) information, making it easy to see pump information, remote control therapy and glucose details - as we move through the day.

It's easy to securely share real-time treatment data with care partner, who can remotely view real-time glucose data and receive un-cleared pump and CGM alerts. It also provides sensor glucose information to better understand glucose levels, and review history to see when levels went above or below target range.

FOTA® App (Firmware-over-the-air):

https://play.google.com/store/apps/details?id=com.medtronic.diabetes.fota&hl=en_AU&gl=au

The FOTA ® mobile app serves as a bridge and a mediator between Server and BLE insulin pump, enabling the insulin pump user to update their insulin pump with a latest firmware. As insulin pump is not internet-enabled but

only BLE enabled, FOTA App communicate with the web server for latest firmware and stream back to insulin pump via BLE.

- ➔ Designed and Created App from scratch with minimal third-party libraries dependency.
- ➔ Utilized kotlin as code based along with use-case design pattern with MVVM as an Architect
- ➔ Implemented Biometric authentication utilizing keystore to signed remote bolus commands
- ➔ Implement security architect for remote bolus signing process for both iOS and Android.
- ➔ Utilized whiteCryption tool and rootbeer library to protect app running on rooted phone and protect app from been tampered.
- ➔ Designed architect for mobile app by utilizing SOLID principles.
- ➔ Designed and Created DTO, POJO, and BO for ease of data transfer between the layers.
- ➔ Designed and Create UI layer using MVVM design pattern.
- ➔ Designed architect to authenticated mobile app with the server by utilizing SafetyNet API.
- ➔ Utilized Adobe Analytic and Google Firebase Crashlytic for app analysis.
- ➔ Utilized Navigation component to navigate across, into, and back out from different screens.
- ➔ Customized L2CAP protocol such as MTU and Connection interval
- ➔ Customized SIG Profiles with custom services, custom characteristics and custom directives.
- ➔ Utilized Mockito library for mocking object for unit testing and as well performed Integration testing.
- ➔ Utilized Espresso framework for user interface/activities white box testing.
- ➔ Created Threat Models, Common Vulnerability Scoring System (CVSS) score and Risk Assessment

SENIOR ANDROID DEVELOPER

March '17– June '19

Medtronic Inc – Northridge, CA

Project:

MiniMed® Connect Pump & CGM App:

https://play.google.com/store/apps/details?id=com.medtronic.diabetes.minimedmobile.us&hl=en_US&gl=US

The MiniMed® Connect app connects to insulin pump via BLE to display key pump and continuous glucose monitor (CGM) information, making it easy to see pump and glucose details - as we move through the day. It's easy to securely shares real-time treatment data with care partner, who can remotely view real-time glucose data and receive un-cleared pump and CGM alerts. It also provides sensor glucose information to better understand glucose levels, and review history to see when levels went above or below target range.

- ➔ Created Multiple Libraries with Inner source development methodology.
- ➔ Created App with Modular and Monolithic architect design.
- ➔ Created Threat Models, Common Vulnerability Scoring System (CVSS) score and Risk Assessment.
- ➔ Created Multiple solutions to fill up security gaps on the mobile app for both iOS and Android.
- ➔ Refactored legacy code to Model View Presenter (MVP) design pattern along with SOLID principles Software design.
- ➔ Created state machines and lightweight state machine-based workflows by using Stateless4j library.
- ➔ Implemented multiple BLE (Smart Bluetooth) Standard Services and custom services.
- ➔ Customized SIG Profiles with custom services, custom characteristics and custom directives.
- ➔ Created BLE GAP advertisement packets for supporting Peripheral mode along with GATT server capability.
- ➔ Implemented customized BLE service for secure communication
- ➔ Participated in different phases of the project cycle such as design, development and testing.
- ➔ Participated in creating software requirement document, software design document, technical document and UI wireframes.
- ➔ Participated in architecting application in multi-tier (multi-layer) like Data, Business logic and UI.
- ➔ Partially migrated Java code based to Kotlin code base with the mixture of both the languages.
- ➔ Utilized Android Jetpack components such as Lifecycleaware, LiveData, Viewmodel, Room etc.
- ➔ Integrated Dagger2 framework for transaction management among different layers of the application.
- ➔ Simplified the complexity of handling threads and asynchronous call by utilizing Rxjava2 library.
- ➔ Implemented Lambda expression by using supporting library Retrolambda in development.
- ➔ Secured SQLite database by implementing SQLCipher library and storing symmetric key in Keystore.
- ➔ Utilized DBFlow library to use ORM approach based on Android native SQLite database.
- ➔ Created DTO, POJO, and BO for ease of data transfer between the layers.

- Created extensive layout for Localization with RTL support and Created library to capture screenshots in all different languages.
- Used Retrofit Library for Type-safe network communication.
- Utilized RenderScript framework for data parallel computation.
- Utilized Mockito library for mocking object for unit testing and as well performed Integration testing.
- Utilized Espresso framework for user interface/activities white box testing.
- Reviewed project for PHI, HIPPA & COPPA Compliance.
- Created, reviewed and ensured regulatory compliance of report files submitted to the FDA

SENIOR SOFTWARE ENGINEER-FULL STACK

Nov '14 – Aug '15

Addrek Smart Solutions – Doha, Qatar

Projects: 1. MyThings App

2. Home Automation System

- Responsible for implementing software solutions utilizing a combination of C++, C#, PHP, Python, Java, JavaScript, and SQL.
- Implement TCP & UDP Protocol for live stream communication.
- Implemented mDNS protocol for searching devices in network, NSD for android and Bonjour for iOS.
- Configuring and implementing Web servers (APACHE), securing web server using SSL/TLS certificate and virtual hosting.
- Working knowledge of a wide range of data center technologies including: local (LAN) and wide-area (WAN); NAS and / or SAN storage technologies. Oracle Technologies, Microsoft windows server, firewalls, and VPNs Network protocols such as TCP/IP, DNS, FTP, HTTP, SMTP.
- Designed the website utilizing Service-oriented MVC architecture, RESTful web services, Python, Angularjs, jQuery, Javascript, CSS and HTML. The client/server communication is TLS/SSL encrypted.
- Developed Mobile App For iOS/Android
- Worked with the Android SDK, and implemented Android BLE (Smart Bluetooth) and Location Connectivity.
- Used web views, list views & populated lists to display the lists from database using simple adapters.
- Used services, Async tasks and Alert Notification API to implement event reminder feature into the mobile app.
- Utilized Zxing library for multi-format 1D/2D barcode image processing library with help of camera.
- Worked on custom Linux operating system using Qt ARM Embedded Linux.
- Worked closely on with embedded team for design and prototype of devices.
- Embedded Google Maps API, GPS Location Data into app with zooming controls.
- Involved in working with Location Manager, Location based services and RESTful Web service with JSON.
- Worked with SQLite databases on Android and IOS and testing of SQL database.
- Responsible for working on networking, security, memory management.
- Implemented unique designs for portrait and landscape mode and support different screen resolutions.

FULL STACK-SOFTWARE ENGINEER

May '13 – Oct '14

BlueLynx IT solutions – Doha, Qatar

Projects: 1. Restaurant Management App

2. Property Manager App

3. BlueLynx Online Store

4. Cheque Management App

- Participated and managed Web Development projects with the following technologies: JavaScript, jQuery, PHP, AJAX, Java, ASP.NET, Perl, HTML, D-HTML languages Flash, Flash Generator, Shockwave, Oracle, SQL Server, Progress databases.
- Developed web applications and websites primarily in PHP/MySQL, utilizing my skill set of HTML, CSS, JavaScript, etc.
- Assisted in product development projects and Integrated UI with large backend applications.
- Aided in writing technical documents for website enhancements and updates.

- ➔ Worked with Architects, Engineering Managers, Accountants and Business Management teams to drive efficiencies and pursue their required mobile apps.
- ➔ Utilized XCode, Swift, Objective-C, Visual Studio, C#, VB .Net, Git, Microsoft SQL Server.
- ➔ Developed Objective-C layer for iOS SDK and JAVA layer for Android SDK.
- ➔ Developed application for unit testing in JAVA and Objective-C.
- ➔ Ensured that all components: resources, representations, versioning, extensibility, security, monitoring, and throttling are kept check.

ORACLE DATABASE ADMINISTRATOR (INTERN)

May '12 – April '13

IIJT Solutions - Hyderabad, India

- ➔ Performed database administration functions in support of the IOC/WK system.
- ➔ Performed database backups and recovery; conducted periodic review and analysis of database fault logs.
- ➔ Created and updated existing database schema's, modified database procedures and functions, created and modified database views, maintained and archived data and executed and diagnosed database scripts.
- ➔ Reviewed and provided input regarding recommended changes for database design. Data Security, Backup & Recovery.
- ➔ Standby/Failover Administration, Oracle RAC & ASM Administration/Installation, Oracle Forms and Reports Development, Database-Backed Web Solutions, SQL Tuning in an Oracle Environment.
- ➔ Experienced in Oracle software installation, upgrades and, migrations from 8i/9i/10g and 11g.
- ➔ Actively participated in workshops for establishing database standards, procedures and, backup strategy.
- ➔ Performed in-depth Oracle internal troubleshooting on high volume transaction on Very Large Databases (VLDB).
- ➔ Created complex Oracle and SQL Reports, Oracle Triggers Procedures and Functions in PL/SQL.

EDUCATION

UNIVERISTY OF THE CUMBERLANDS

May 2020

Executive master's in information systems security

NORTHWESTERN UNIVERISTY

Dec 2016

Master's in computer science